LEARNING UNIT 4

Operations Management in a Digital World

Transforming operational workflows through Lean-Agile methodologies, automation, digital supply chains, customer journey optimization, and real-time dashboard analytics.

- Delivery Duration: 3 Hours (180 Mins)
- Mid-Session Break: 15 Mins
- 4 Core Learning Outcomes

Session Schedule & Lesson Timings

Total Duration: 3h 15m

Time Frame	Module / Focus	Learning Content & Core Activities	Delivery Method
00:00 - 00:15 (15m)	**Intro & Setup**	Course overview, learning outcomes alignment, and context setting.	Interactive Lecture
00:15 - 00:55 (40m)	**LO1: Lean & Agile**	Process waste elimination, responsiveness + **Simulation Workshop** (25m)	Theory + Group Activity
00:55 - 01:35 (40m)	**LO2: Digital Chains**	RPA, automated inventory, supply chain + **Evaluation Case Study** (25m)	Theory + Case Study
01:35 - 01:50 (15m)	**NETWORKING BREAK**	15-Minute Refreshment & Informal Discussion Break	Break Period
01:50 - 02:30 (40m)	**LO3: Service Design**	Customer journey mapping, digital touchpoints + **Journey Mapping Lab** (25m)	Theory + Mapping Lab
02:30 - 03:10 (40m)	**LO4: KPIs & Dashboards**	Operational metrics, real-time KPI design + **Dashboard Studio** (25m)	Theory + Hands-on Lab
03:10 - 03:25 (15m)	**Wrap-Up & Q&A**	Key takeaway synthesis, workplace implementation plan, and open Q&A.	Class Discussion

Overall Unit Learning Outcomes

Unit Framework

LEARNING OUTCOME 1

Agile & Lean Operations

Apply Agile and Lean operational principles to eliminate process waste and increase organizational responsiveness in digital workflows.

LEARNING OUTCOME 2

Digital Supply Chains & RPA

Evaluate end-to-end digital supply chains, Robotic Process Automation (RPA), and real-time inventory management systems.

LEARNING OUTCOME 3

Customer Journey & Service Design

Map customer journeys and apply service design methodologies to optimize digital touchpoints and elevate user experience.

LEARNING OUTCOME 4

Operational KPIs & Dashboards

Establish operational KPIs and digital dashboard metrics to evaluate process performance and continuous operational efficiency.

LO1: Agile & Lean Principles

15 Mins Concept

Eliminating Waste & Increasing Agility

In digital operations, Lean focuses on eliminating non-value-adding steps (Muda), while Agile enables rapid response to shifting market conditions.

- **Identify 8 Digital Wastes:** Defects, overprocessing, waiting times, unused talent, and unnecessary data movement.
- **Iterative Sprints:** Breaking operational transformation into rapid 2-week execution cycles.
- **Pull Systems:** Triggering operational tasks based on real-time customer demand signals.
- **Continuous Improvement (Kaizen):** Empowering cross-functional teams to streamline daily operational flows.

LO1 Activity: Process Waste Lab

25 Mins Interactive

TEAM WORKSHOP (GROUPS OF 3-4)

Lean Process Waste Audit & Redesign

Scenario: A digital order fulfillment process suffers from 40% lead time delays due to manual approvals and redundant data entry.

Activity Instructions:

1. Review the sample order workflow diagram provided.
2. Identify at least 3 forms of digital waste (e.g., waiting, overprocessing).
3. Apply Lean-Agile principles to eliminate 2 redundant steps.
4. Draft a revised, streamlined workflow structure.

Activity Breakdown (25 Mins)

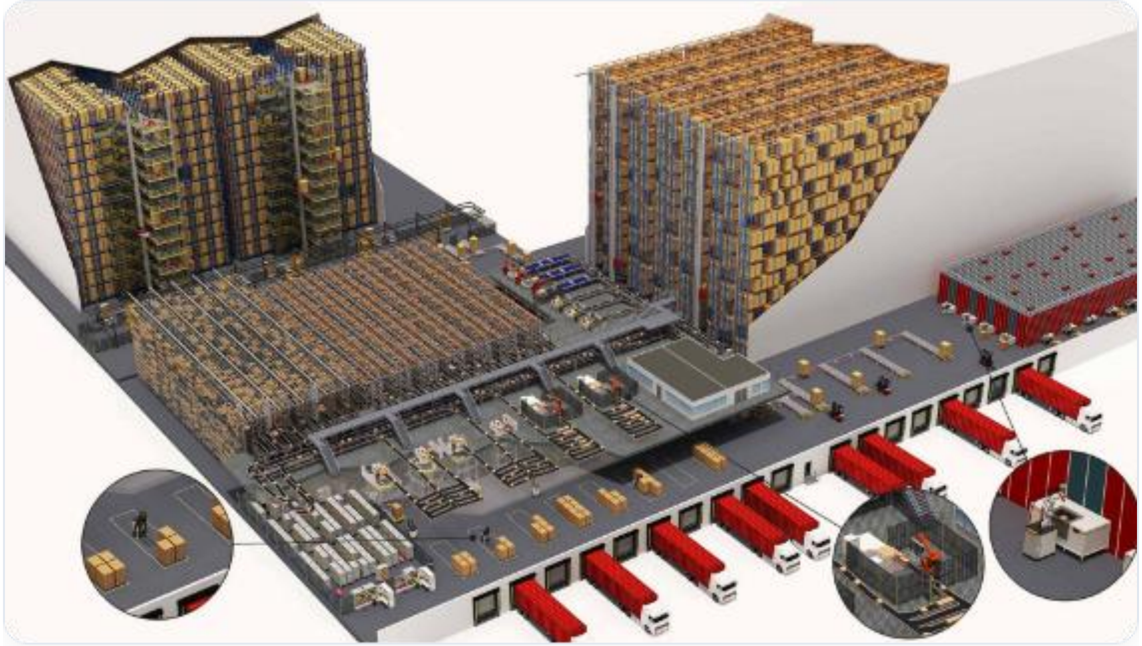
- **05 Mins:** Team setup & scenario briefing
- **12 Mins:** Waste identification & workflow redesign
- **08 Mins:** Group presentations & feedback discussion

Expected Deliverable

A 1-page "Before vs. After" Lean workflow diagram showing eliminated bottlenecks and calculated cycle-time savings.

LO2: Digital Supply Chains & RPA

15 Mins Concept



Automating End-to-End Operations

Integrating Robotic Process Automation (RPA) and real-time IoT tracking creates a seamless, error-free supply chain architecture.

- **Robotic Process Automation (RPA):** Automates repetitive transactional tasks such as invoice matching and order status updates.
- **Real-Time Inventory Systems:** RFID and IoT sensor integration for precise stock visibility across all fulfillment nodes.
- **Connected Supply Ecosystems:** Direct API connectivity between suppliers, logistics providers, and end customers.

LO2 Activity: RPA Feasibility Case Study

25 Mins Interactive

CASE STUDY ANALYSIS

RPA & Inventory Automation Matrix

Scenario: Global Logistics Corp experiences high error rates in manual inventory reconciliation across 3 regional distribution hubs.

Activity Tasks:

1. Evaluate 4 candidate tasks for RPA implementation using the Feasibility Matrix.
2. Select top 2 tasks with highest ROI and lowest implementation complexity.
3. Propose a real-time inventory tracking integration model.

TASK 1: EVALUATION (10 MINS)

Assess rule-based tasks vs. exception-heavy tasks using candidate matrix.

TASK 2: ROI CALCULATION (08 MINS)

Estimate cost-per-transaction reduction following bot deployment.

TASK 3: DEBRIEF (07 MINS)

Compare team evaluation scores and refine automation criteria.

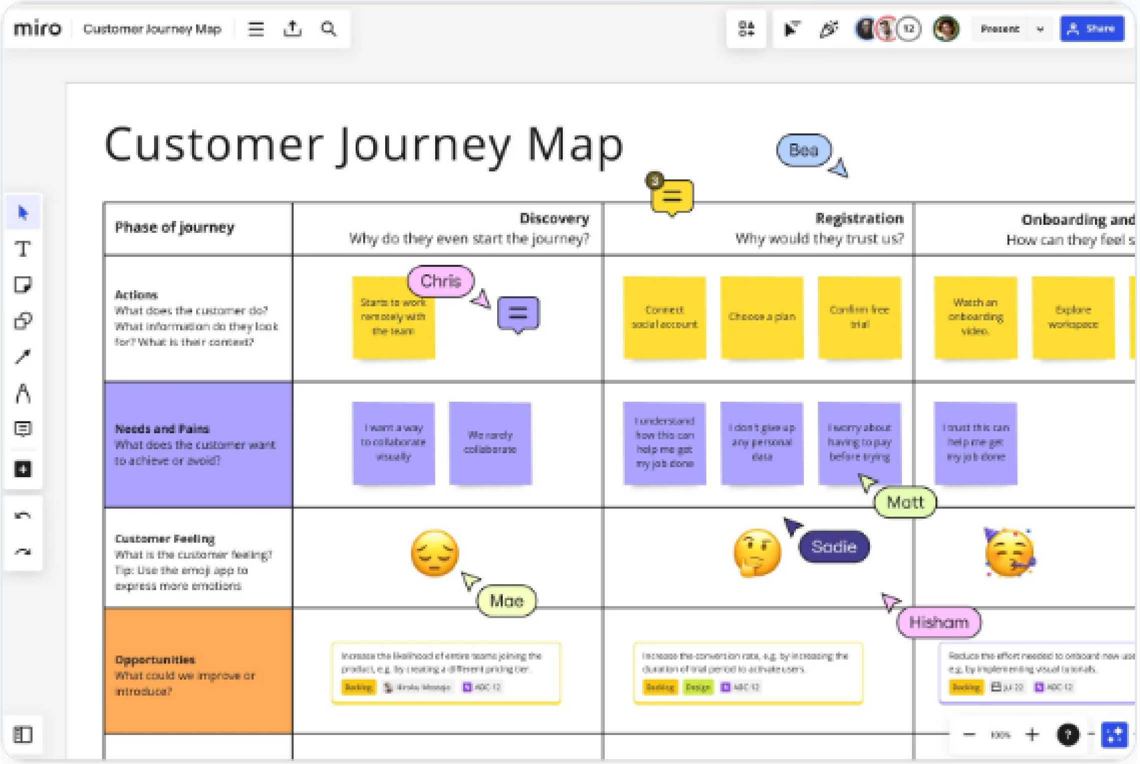
LO3: Customer Journey & Service Design

15 Mins Concept

Service Design & Touchpoint Optimization

Aligning front-stage user experiences with back-stage operational processes ensures seamless customer interactions.

- **Customer Journey Mapping:** Visualizing steps, emotion levels, pain points, and moments of truth.
- **Service Blueprinting:** Connecting customer actions directly to operational support processes and IT infrastructure.
- **Omnichannel Touchpoints:** Creating consistent digital experiences across mobile apps, web, and support channels.



LO3 Activity: Journey Mapping Studio

25 Mins Interactive

HANDS-ON MAPPING

Digital Touchpoint Optimization Exercise

Scenario: Users report a friction-heavy onboarding process for an enterprise SaaS operational platform.

Lab Steps:

1. Map out the 5 key onboarding phases (Awareness -> Signup -> Setup -> Usage -> Support).
2. Identify key digital touchpoints and pin high-friction pain points.
3. Redesign 2 critical touchpoints to improve user satisfaction (CSAT).

Activity Objective

Develop a high-impact **Service Blueprint Draft** that links optimized customer touchpoints directly to backend digital automation systems.

Timing: 15m Mapping | 10m Group Showcase

LO4: KPIs & Digital Dashboards

15 Mins Concept

Data-Driven Operational Decision Making

Transforming raw operational logs into actionable real-time intelligence using executive and operational dashboards.

- **Core Operational KPIs:** Cycle Time, Order Fulfillment Rate (OTIF), First Contact Resolution (FCR), and Resource Utilization.
- **Dashboard Architecture:** Real-time visual metrics, threshold alerts, and drill-down analytical capabilities.

99.4% On-Time In-Full (OTIF)	**12.5m** Avg Cycle Time	**0.02%** Process Error Rate	**94/100** Health Index Score

LO4 Activity: Metric & Dashboard Design

25 Mins Interactive

HANDS-ON METRIC LAB

Executive Dashboard Metric Specification

Task: Design a real-time operational dashboard wireframe for a Chief Operating Officer (COO) overseeing digital operations.

Lab Requirements:

1. Select 4 critical operational KPIs (Efficiency, Quality, Speed, Cost).
2. Define metric calculation formulas and target thresholds for each KPI.
3. Sketch a visual layout structure for key widgets and status alerts.

Sample Formula Definition

Order Fulfillment Rate Formula:

$$\text{OTIF Rate} = \frac{\text{Orders Delivered On-Time \& In-Full}}{\text{Total Orders Processed}} \times 100$$

Time Allocation (25 Mins)

15m Wireframe & KPI Definition | 10m Peer Review

Unit Summary & Action Plan

15 Mins Wrap-Up

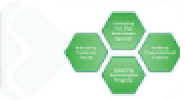
Summary of Key Takeaways

- **LO1:** Lean-Agile principles eliminate waste and boost agility.
- **LO2:** RPA and IoT enable automated, real-time supply chains.
- **LO3:** Service design connects touchpoints to backend processes.
- **LO4:** Real-time KPI dashboards drive data-backed operational decisions.

Questions & Discussion

How will you apply these digital operations concepts to your immediate work environment?

Image Sources



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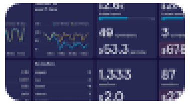
Source: miro.com



https://www.geckoboard.com/uploads/Digital-dashboard-example.png

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